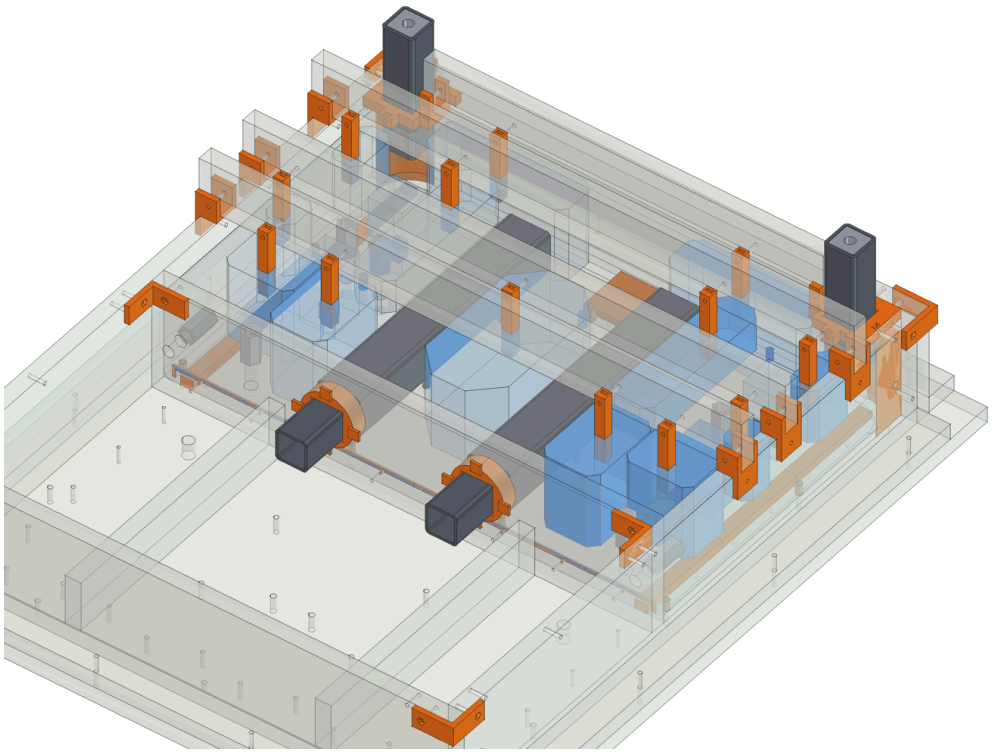
Mold guide

DIY CNC epoxy-granite mill · October 2026

How to make the molds for the five castings, put them together, and take them apart again. The molds are melamine boards that bolt together with printed brackets; every printed piece bolts or pins to its board from outside, and the bulky hidden cores are foam. Nothing is glued, nothing goes through a face the mix touches, and the molds come apart cleanly for the next casting.

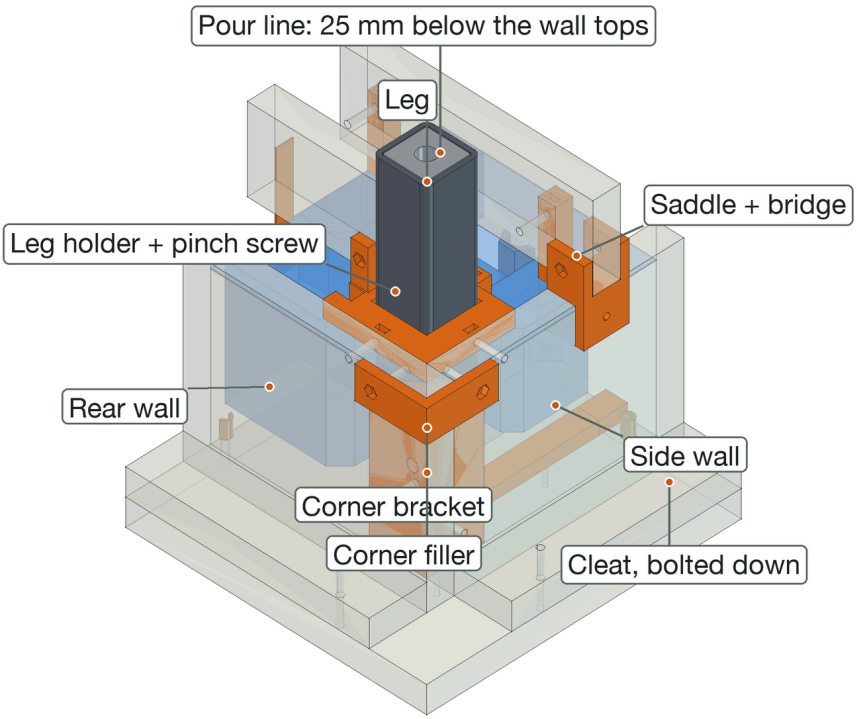
The build guide covers the casting itself (mix, pour, cure); this guide is everything before and after that. All the sizes and hole positions come from the model (`model/build_cnc.py`), so if you change the design, re-run it and this guide, the hole maps and the printed parts all follow.



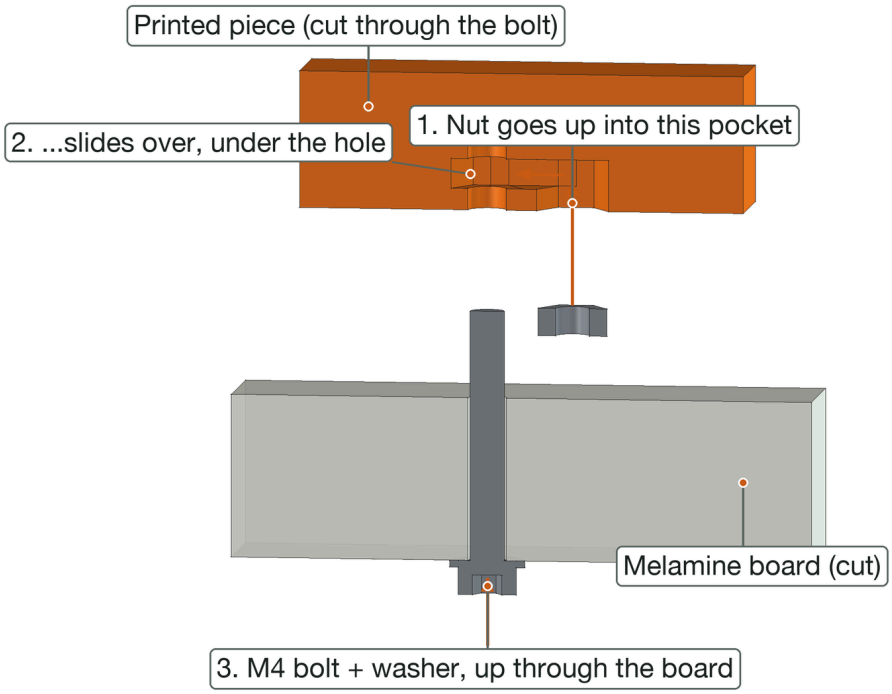
The order of work: print the kit (about a week of printer time, so start first) → cut and drill the boards → cut the foam → wax → assemble a mold → cast → take it apart. Cast in this order: base rear half, base front half (against the rear half), the two uprights together, the beam.

1. How the molds work

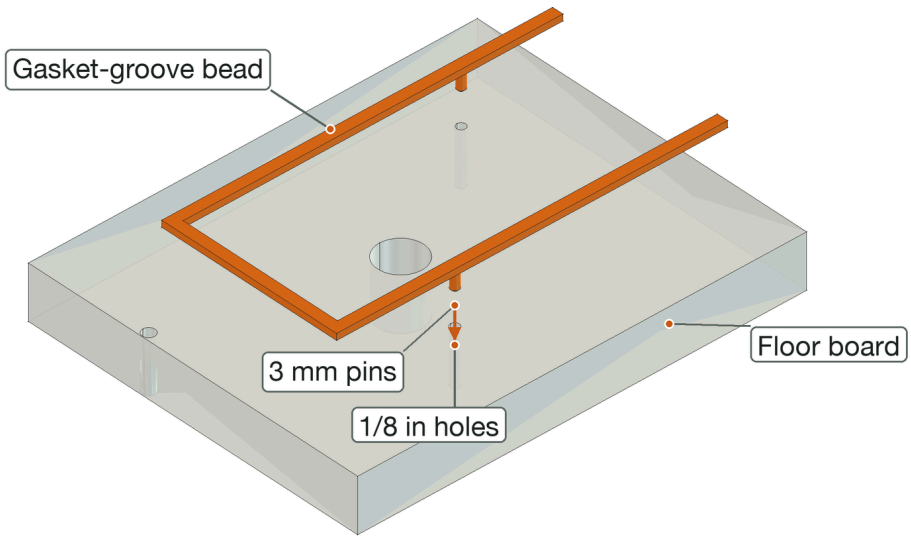
Walls stand 25 mm above the pour line. All the bolts that hold the box together go through that band, so none passes through a face that gets cast against. Fill to the line, not to the top. Strips of board (cleats) bolted down along the outside of each wall's foot keep the walls from moving out; printed corner brackets hold the corners square at the top.



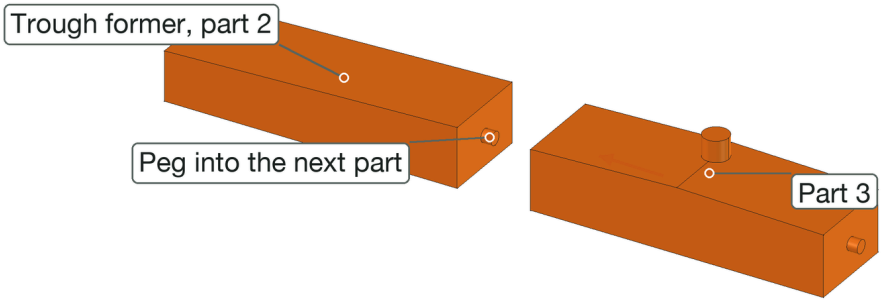
Printed pieces bolt on from outside. Each piece that forms part of the casting's surface (the formers and fillers) has a nut built into it: the nut drops into a pocket in the face that sits against the board, slides sideways under the bolt hole, and an M4 bolt (M3 on thin pieces) goes through the board from outside into it. The pocket is on the board side, so nothing opens onto a cast face, and the bolt comes out from outside after casting. Fit the nuts before the pieces go on.



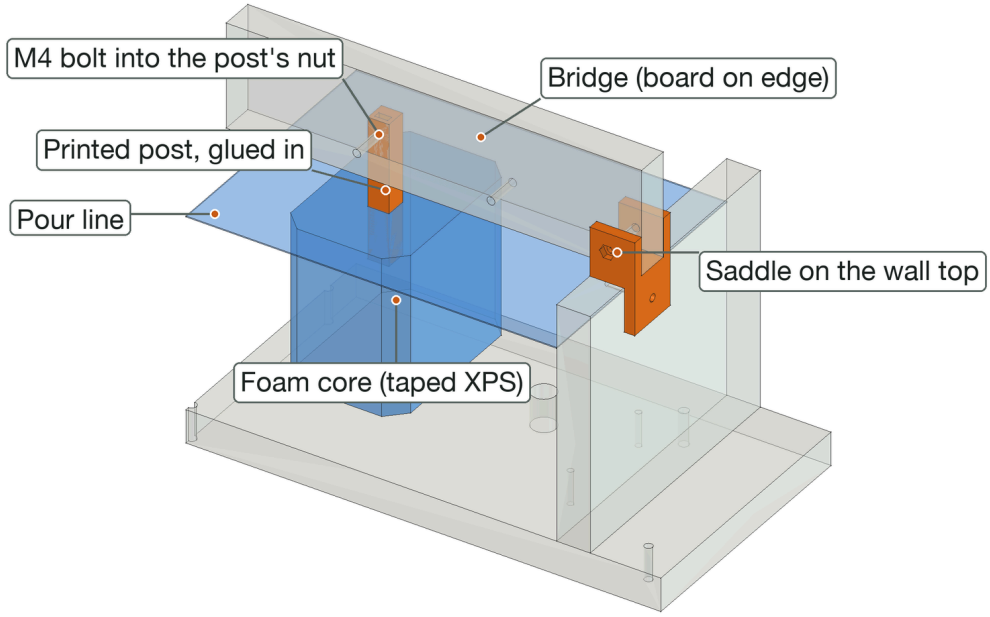
Thin pieces have pins. The gasket-groove beads and the motor relief are too thin for a nut; their 3 mm pins push into 1/8 in holes. The casting presses them onto the board.



Pieces longer than the printer bed are printed in parts with a peg on one cut face and a socket in the other. Push them together; each part has its own bolts to the board, so they need no glue.



Cores (the pockets in the base underside) are blocks of XPS foam wrapped in packing tape. Each has a printed post glued into it; the post bolts to the side of a bridge (a board standing on edge across the mold, in printed saddles on the wall tops), which sets its height and stops it floating.



Fits are 0.15 mm a side: snug, but they go together by hand. Board thickness matters: the saddles and cradles are drawn for 19.05 mm board (Home Depot's 3/4 in white melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 19 mm (3/4 in)	2 sheets, 4 x 8 ft	the layout below; have the store rip the strips
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores; stack two layers for the 90 mm ones
PETG	2 x 1 kg rolls	1.19 kg for the molds and cast-in parts (sliced)
Packing tape, double-sided carpet tape	1 roll each	foam wrapping and holding
Paste wax (Partall #2) and PVA	1 each	release, 3-5 coats
3/8 in quarter-round molding (PVC)	about 6 m	the 10 mm outside edges of the castings
2x4 offcut	4 blocks, about 50 mm	feet for each upright mold

Hardware (zinc-plated is fine; grease anything that touches the mix). Buy about 10 % spare of the small bolts, nuts and washers.

Item	Qty
M3 nut	20

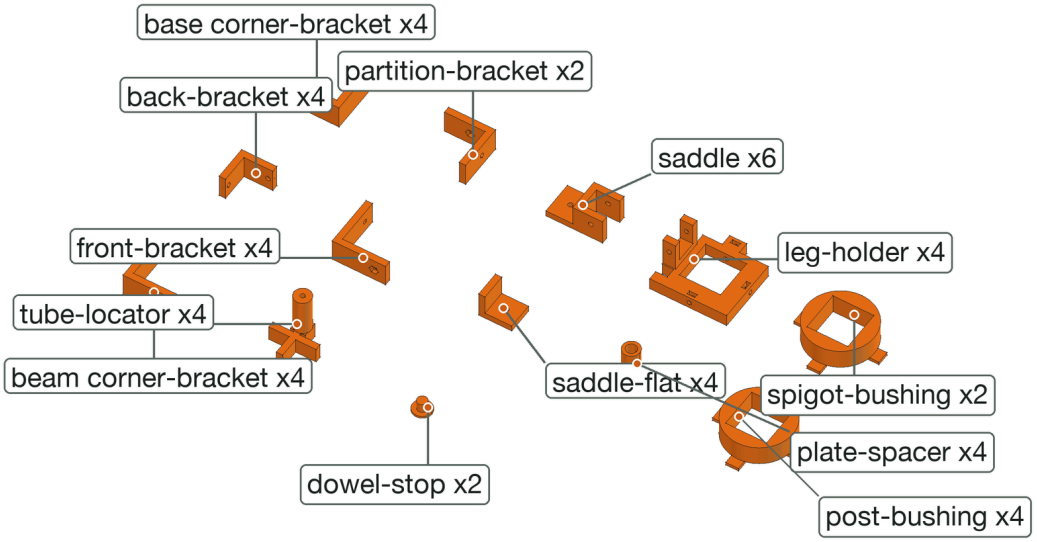
Item	Qty
M3 washer	20
M3 x 25 bolt	20
M4 nut	180
M4 washer	176
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	104
M4 x 35 bolt	28
M4 x 45 bolt	40
M4 x 70 bolt	4
M5 x 20 screw, greased (holds an insert)	29
M6 x 20 screw, greased (holds a pad)	6
M6 x 20 screw, greased (holds an insert)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 198 mm	8
1/2-13 x 1-1/2 in bolt + nut, greased (forms the dowel hole)	2
1/2-13 x 1-1/2 in bolt, greased (holds a draw-rod coupler)	2
1/2-13 x 2-1/2 in bolt, greased (holds a coupler)	8
3/8-16 x 2-1/2 in bolt, greased (holds the top plate)	4
5/8 in plain steel rod, greased (forms a tie-rod hole), cut to 140 mm, 243 mm, 377 mm, 377 mm	2

Tools: a drill with sharp brad-point bits (sizes in section 5) and a 2-1/2 in hole saw; a jigsaw or a circular saw for the upright side plates' sloped edge; a square and a long straightedge; a few clamps; a 7 mm nut driver and a 3 mm hex key; a fine saw or a hot-wire cutter and a sanding block for the foam; a marker that writes on melamine.

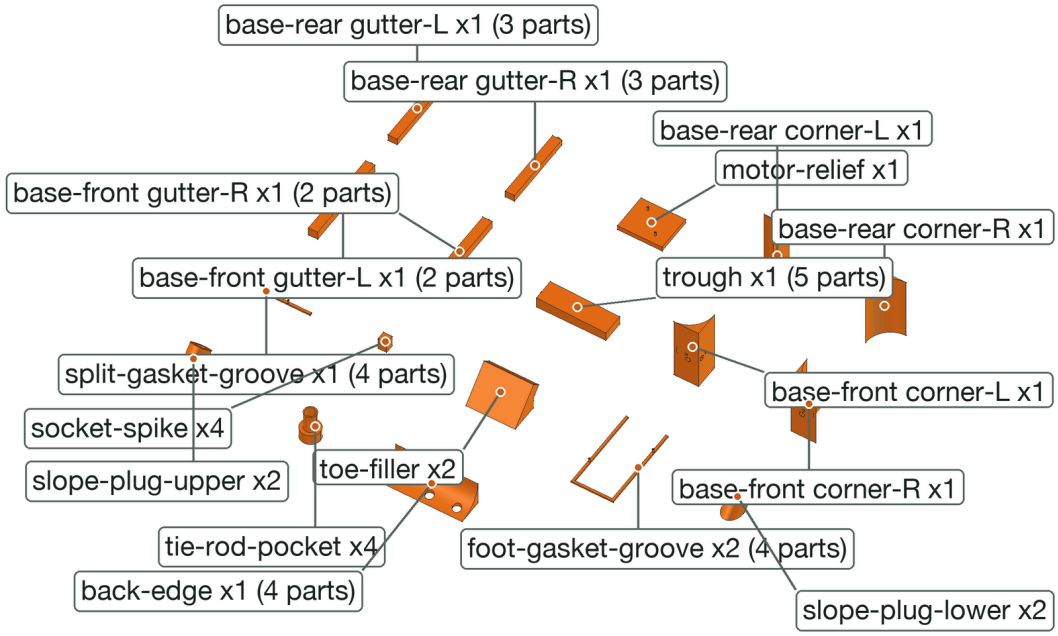
3. Print the kit

Everything printed is in `print/` and listed in the print guide: 144 prints, 1.29 kg of PETG and about 107 hours of printing, sliced in Bambu Studio for an A1 mini. Start early. Print order: one corner bracket and one saddle first (check them on an offcut of your board: the saddle should drop over it with a little play), then the kit, then the pieces for the rear base half, then the rest.

The mold kit (reused for every casting):



The formers, fillers and plugs (each shapes part of a casting):

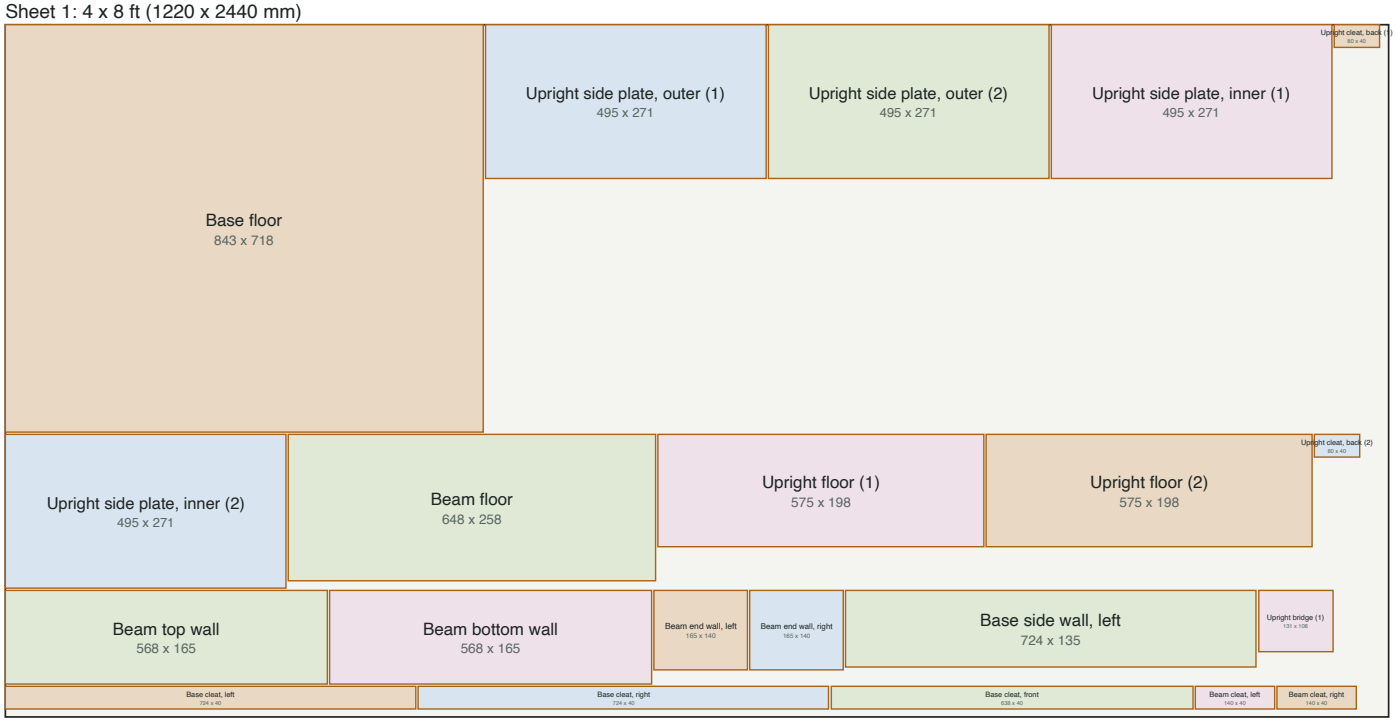


- **Nuts:** press M4 nuts into the hex pockets of the brackets and saddles; slide the nuts into the pockets of the formers and fillers (a dab of grease keeps them from falling out). A 7 mm nut on a bolt through the hole pulls a tight one into place.
- **Pegs:** test-fit the parts of each long former; a light pass with a file if a peg is tight.

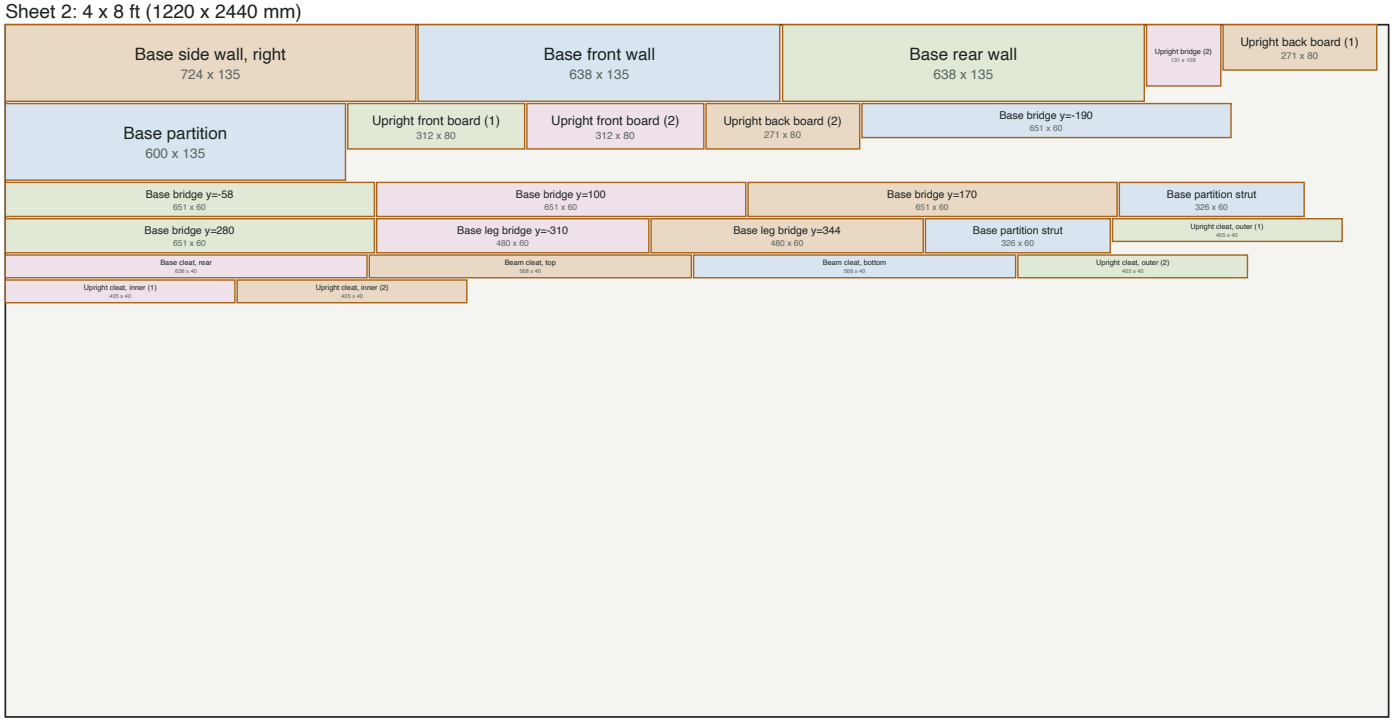
4. Cut the boards

Most boards are strips, so the quickest way is to have the store rip each sheet into strips at the widths in the layout, and crosscut them at home. Label every board with its name on the outside face as you cut it, and mark which face is the cast face (the melamine face that the mix touches).

Sheet 1: 4 x 8 ft (1220 x 2440 mm)



Sheet 2: 4 x 8 ft (1220 x 2440 mm)



Mold	Board	Qty	Size (mm)	Notes
Base	Base floor	1	718 x 843	both halves are cast on it

Mold	Board	Qty	Size (mm)	Notes
Base	Base front wall	1	638 x 135	
Base	Base rear wall	1	638 x 135	
Base	Base side wall, left	1	724 x 135	
Base	Base side wall, right	1	724 x 135	
Base	Base partition	1	600 x 135	rear half only; comes out before the front half
Base	Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base	Base cleat, front	1	638 x 40	offcut strip, flat on the floor
Base	Base cleat, rear	1	638 x 40	offcut strip, flat on the floor
Base	Base cleat, left	1	724 x 40	offcut strip, flat on the floor
Base	Base cleat, right	1	724 x 40	offcut strip, flat on the floor
Base	Base bridge y=-190	1	651 x 60	
Base	Base bridge y=-58	1	651 x 60	
Base	Base bridge y=100	1	651 x 60	
Base	Base bridge y=170	1	651 x 60	
Base	Base bridge y=280	1	651 x 60	
Base	Base leg bridge y=-310	1	480 x 60	
Base	Base leg bridge y=344	1	480 x 60	
Uprights	Upright floor	2	198 x 575	
Uprights	Upright side plate, outer	2	495 x 271 (profile)	cut to the profile
Uprights	Upright side plate, inner	2	495 x 271 (profile)	cut to the profile
Uprights	Upright back board	2	80 x 271	
Uprights	Upright front board	2	80 x 312	ends cut square
Uprights	Upright bridge	2	131 x 108	
Uprights	Upright cleat, outer	2	405 x 40	offcut strip, flat on the floor
Uprights	Upright cleat, inner	2	405 x 40	offcut strip, flat on the floor
Uprights	Upright cleat, back	2	80 x 40	offcut strip, flat on the floor
Beam	Beam floor	1	648 x 258	
Beam	Beam top wall	1	568 x 165	
Beam	Beam bottom wall	1	568 x 165	
Beam	Beam end wall, left	1	140 x 165	
Beam	Beam end wall, right	1	140 x 165	
Beam	Beam cleat, top	1	568 x 40	offcut strip, flat on the floor
Beam	Beam cleat, bottom	1	568 x 40	offcut strip, flat on the floor
Beam	Beam cleat, left	1	140 x 40	offcut strip, flat on the floor
Beam	Beam cleat, right	1	140 x 40	offcut strip, flat on the floor

The two upright side plates are cut to a profile (their drawings give the corners): cut the rectangle, then the sloped front edge. The front board's ends are square.

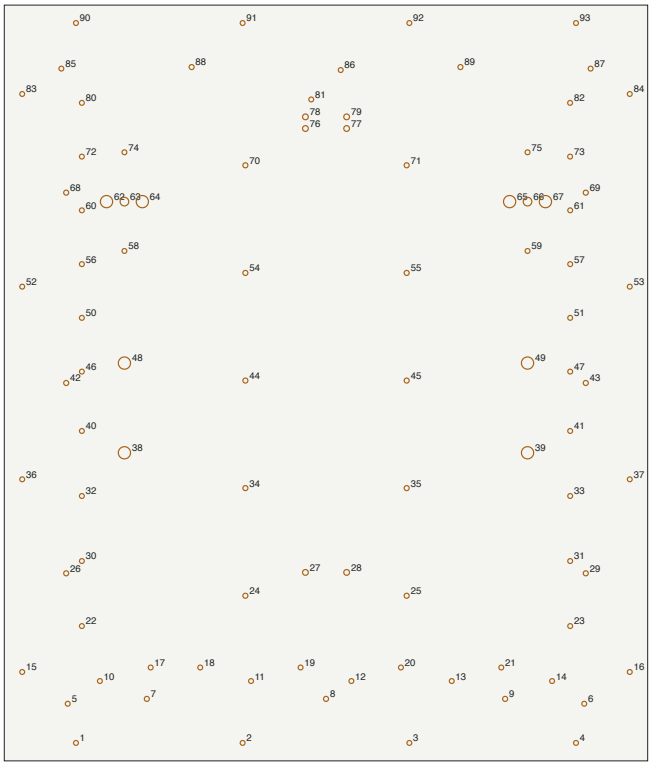
5. Drill the boards

- Drill every hole from the **cast side**, so any tear-out is on the outside. Clamp a scrap under the board where the bit comes out.
- The drawings and tables are seen from the cast side: x from the left edge, y from the bottom edge. For a wall the bottom edge sits on the floor; for a floor, the bottom of the drawing is toward the front wall (the beam's bottom wall). Left and right are as you look at the cast face, so a floor is mirrored compared with the machine. Bridges and cleats are drawn from the top.
- The brackets and saddles are drill guides: clamp one in place and drill through its holes.
- Countersink nothing; a washer goes under every bolt head.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove, motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	back-edge, cleat, core post, corner bracket, corner-L, corner-R, gutter-L, gutter-R, leg holder, saddle, slope-plug-lower, slope-plug-upper, socket spike, toe-filler, trough, tube locator
5.5 mm	7/32 in	M5 insert screw
6.5 mm	17/64 in	M6 insert screw
9.6 mm	25/64 in	3/8 dowel
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2 core rod, 1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	1/2-13 bolt into coupler
16.3 mm	41/64 in	5/8 core rod
63.5 mm	2-1/2 in hole saw	hole saw

Base

Base floor



0,0 · 718 x 843 mm · seen from the cast side

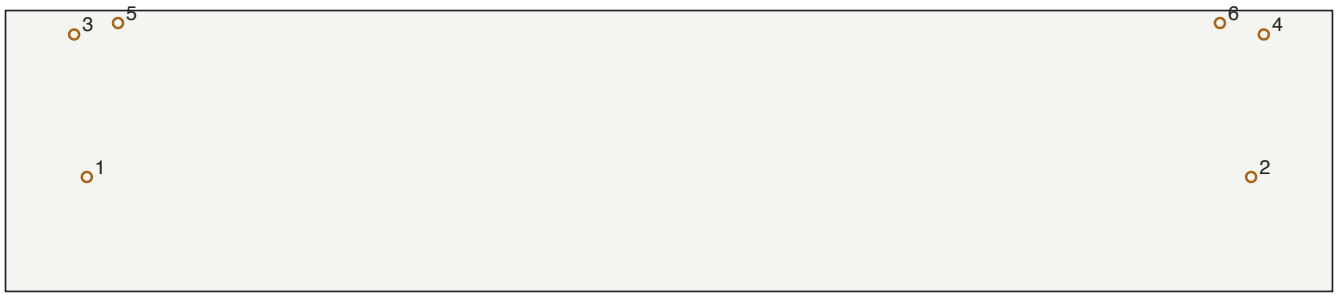
#	x	y	Hole	For
1	80	20	4.5	cleat
2	266	20	4.5	cleat
3	452	20	4.5	cleat
4	638	20	4.5	cleat
5	71	64	3.4	corner-R
6	647	64	3.4	corner-L
7	159	69	5.5	M5 insert screw
8	359	69	5.5	M5 insert screw
9	559	69	5.5	M5 insert screw
10	107	89	4.5	trough
11	275	89	4.5	trough
12	387	89	4.5	trough
13	499	89	4.5	trough
14	611	89	4.5	trough
15	20	99	4.5	cleat
16	698	99	4.5	cleat
17	163	104	4.5	trough
18	219	104	4.5	trough
19	331	104	4.5	trough

#	x	y	Hole	For
20	443	104	4.5	trough
21	555	104	4.5	trough
22	87	150	4.5	gutter-R
23	632	150	4.5	gutter-L
24	269	184	5.5	M5 insert screw
25	449	184	5.5	M5 insert screw
26	69	209	5.5	M5 insert screw
27	336	210	6.5	M6 insert screw
28	382	210	6.5	M6 insert screw
29	649	209	5.5	M5 insert screw
30	87	223	4.5	gutter-R
31	632	223	4.5	gutter-L
32	87	295	3.4	gutter-R
33	632	295	3.4	gutter-L
34	269	304	5.5	M5 insert screw
35	449	304	5.5	M5 insert screw
36	20	314	4.5	cleat
37	698	314	4.5	cleat
38	134	344	13.5	1/2-13 bolt into coupler
39	584	344	13.5	1/2-13 bolt into coupler
40	87	368	3.4	gutter-R
41	632	368	3.4	gutter-L
42	69	421	5.5	M5 insert screw
43	649	421	5.5	M5 insert screw
44	269	424	5.5	M5 insert screw
45	449	424	5.5	M5 insert screw
46	87	434	3.4	gutter-R
47	632	434	3.4	gutter-L
48	134	444	13.5	1/2-13 bolt into coupler
49	584	444	13.5	1/2-13 bolt into coupler
50	87	494	3.4	gutter-R
51	632	494	3.4	gutter-L
52	20	529	4.5	cleat
53	698	529	4.5	cleat
54	269	544	5.5	M5 insert screw

#	x	y	Hole	For
55	449	544	5.5	M5 insert screw
56	87	554	3.4	gutter-R
57	632	554	3.4	gutter-L
58	134	569	4.5	socket spike
59	584	569	4.5	socket spike
60	87	614	3.4	gutter-R
61	632	614	3.4	gutter-L
62	114	624	13.5	1/2-13 bolt into coupler
63	134	624	9.6	3/8 dowel
64	154	624	13.5	1/2-13 bolt into coupler
65	564	624	13.5	1/2-13 bolt into coupler
66	584	624	9.6	3/8 dowel
67	604	624	13.5	1/2-13 bolt into coupler
68	69	634	5.5	M5 insert screw
69	649	634	5.5	M5 insert screw
70	269	664	5.5	M5 insert screw
71	449	664	5.5	M5 insert screw
72	87	674	3.4	gutter-R
73	632	674	3.4	gutter-L
74	134	679	4.5	socket spike
75	584	679	4.5	socket spike
76	336	705	6.5	M6 insert screw
77	382	705	6.5	M6 insert screw
78	336	718	6.5	M6 insert screw
79	382	718	6.5	M6 insert screw
80	87	734	3.4	gutter-R
81	343	737	3.2	motor-relief
82	632	734	3.4	gutter-L
83	20	744	4.5	cleat
84	698	744	4.5	cleat
85	64	772	3.4	corner-R
86	375	770	3.2	motor-relief
87	654	772	3.4	corner-L
88	209	774	5.5	M5 insert screw
89	509	774	5.5	M5 insert screw

#	x	y	Hole	For
90	80	823	4.5	cleat
91	266	823	4.5	cleat
92	452	823	4.5	cleat
93	638	823	4.5	cleat

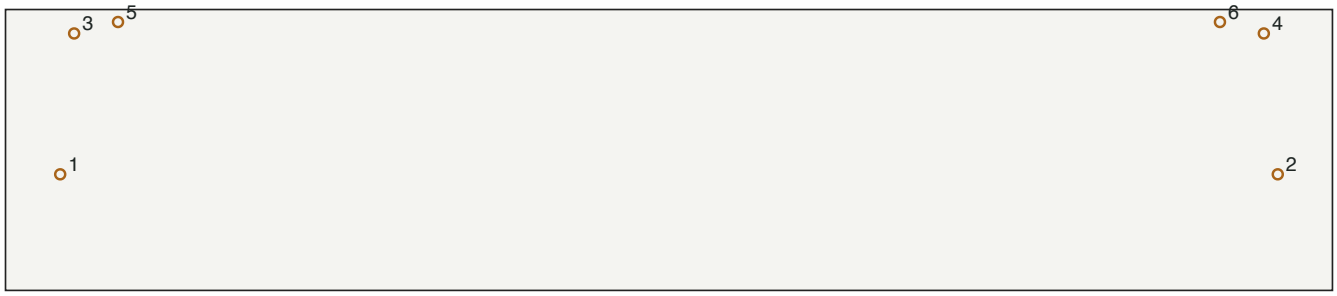
Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	39	55	4.5	corner-L
2	599	55	4.5	corner-R
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

Base rear wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	26	56	4.5	corner-R
2	612	56	4.5	corner-L
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket

#	x	y	Hole	For
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

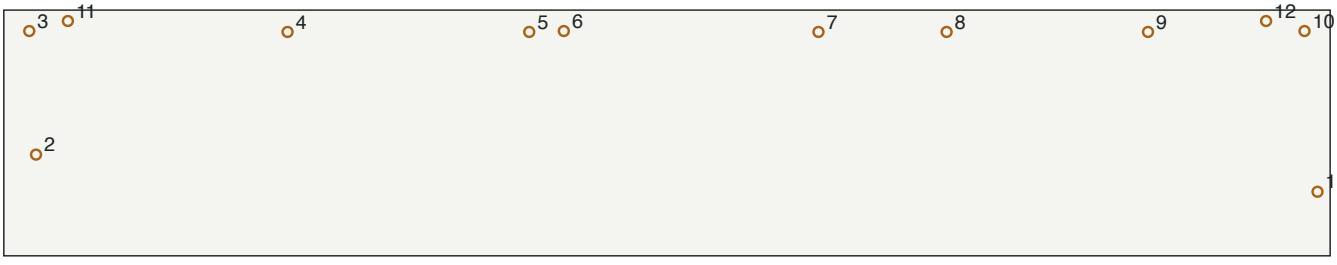
Base side wall, left



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	707	54	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	438	123	4.5	saddle
9	570	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base side wall, right

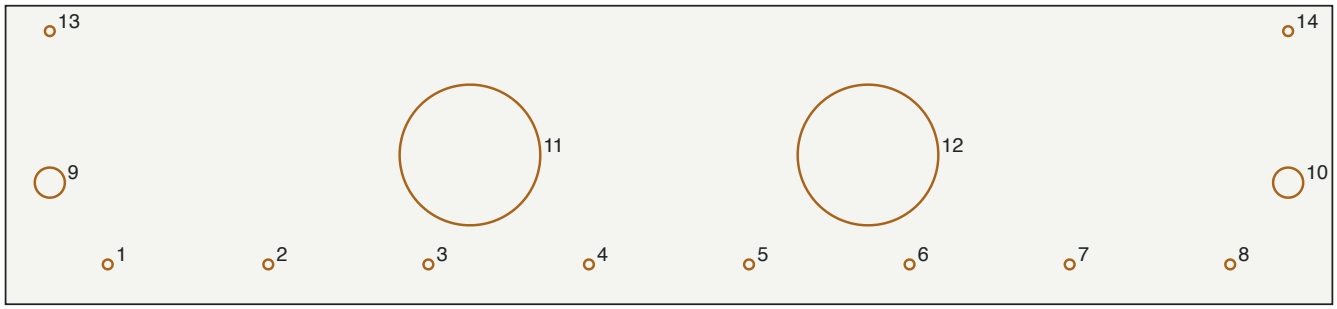


0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	718	35	4.5	corner-R

#	x	y	Hole	For
2	18	56	4.5	corner-R
3	14	124	4.5	corner bracket
4	155	123	4.5	saddle
5	287	123	4.5	saddle
6	306	124	4.5	corner bracket
7	445	123	4.5	saddle
8	515	123	4.5	saddle
9	625	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base partition

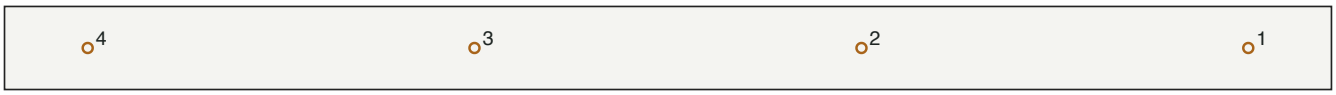


0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	20	55	13.5	1/2-13 bolt into coupler
10	580	55	13.5	1/2-13 bolt into coupler
11	210	68	63.5	hole saw
12	390	68	63.5	hole saw
13	20	124	4.5	corner bracket

#	x	y	Hole	For
14	580	124	4.5	corner bracket

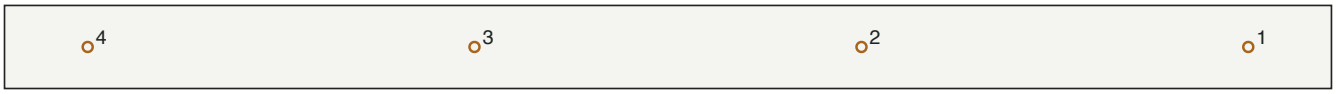
Base cleat, front



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

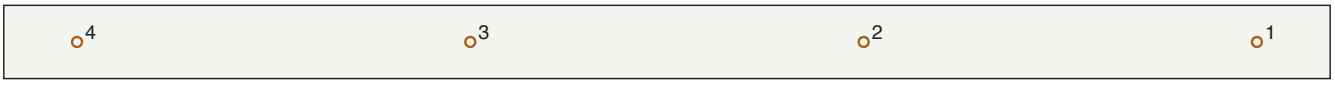
Base cleat, rear



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

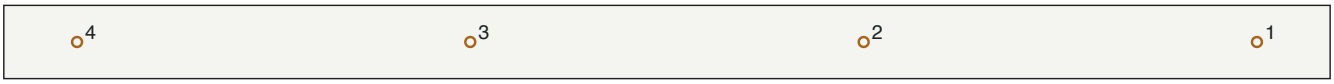
Base cleat, left



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

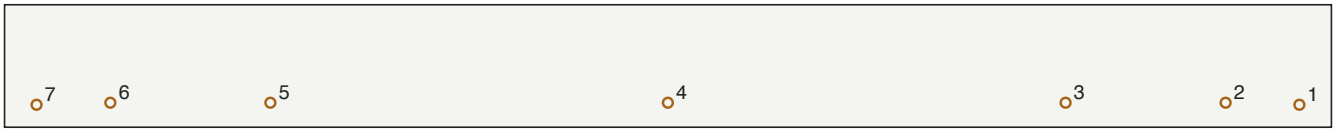
Base cleat, right



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

Base bridge y=-190



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	599	12	4.5	core post
3	520	12	4.5	core post
4	325	12	4.5	core post
5	130	12	4.5	core post
6	52	12	4.5	core post
7	16	11	4.5	saddle

Base bridge y=-58



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	489	12	4.5	core post
3	161	12	4.5	core post
4	16	11	4.5	saddle

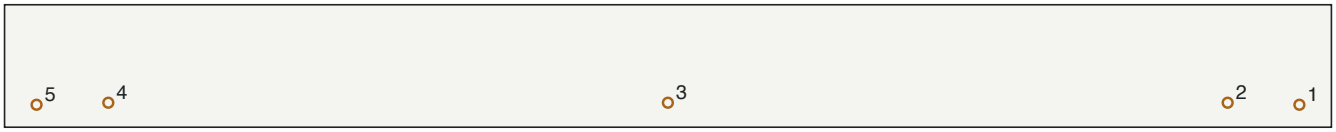
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	565	12	4.5	core post
3	489	12	4.5	core post
4	161	12	4.5	core post
5	85	12	4.5	core post
6	16	11	4.5	saddle

Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	325	12	4.5	core post
4	51	12	4.5	core post
5	16	11	4.5	saddle

Base bridge y=280

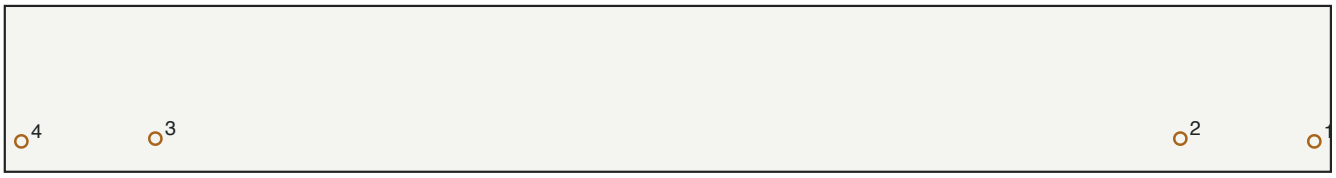


0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	480	12	4.5	core post
4	170	12	4.5	core post

#	x	y	Hole	For
5	51	12	4.5	core post
6	16	11	4.5	saddle

Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344

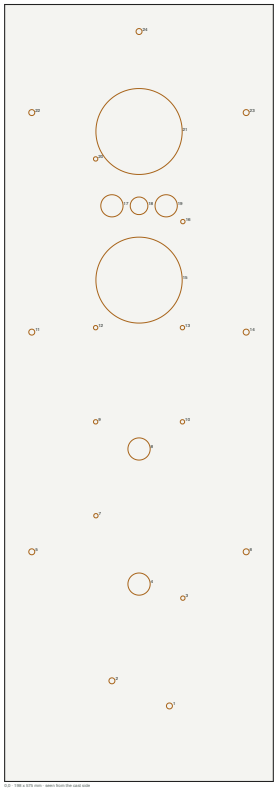


0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	385	12	4.5	core post
3	95	12	4.5	core post
4	6	11	4.5	leg holder

Uprights

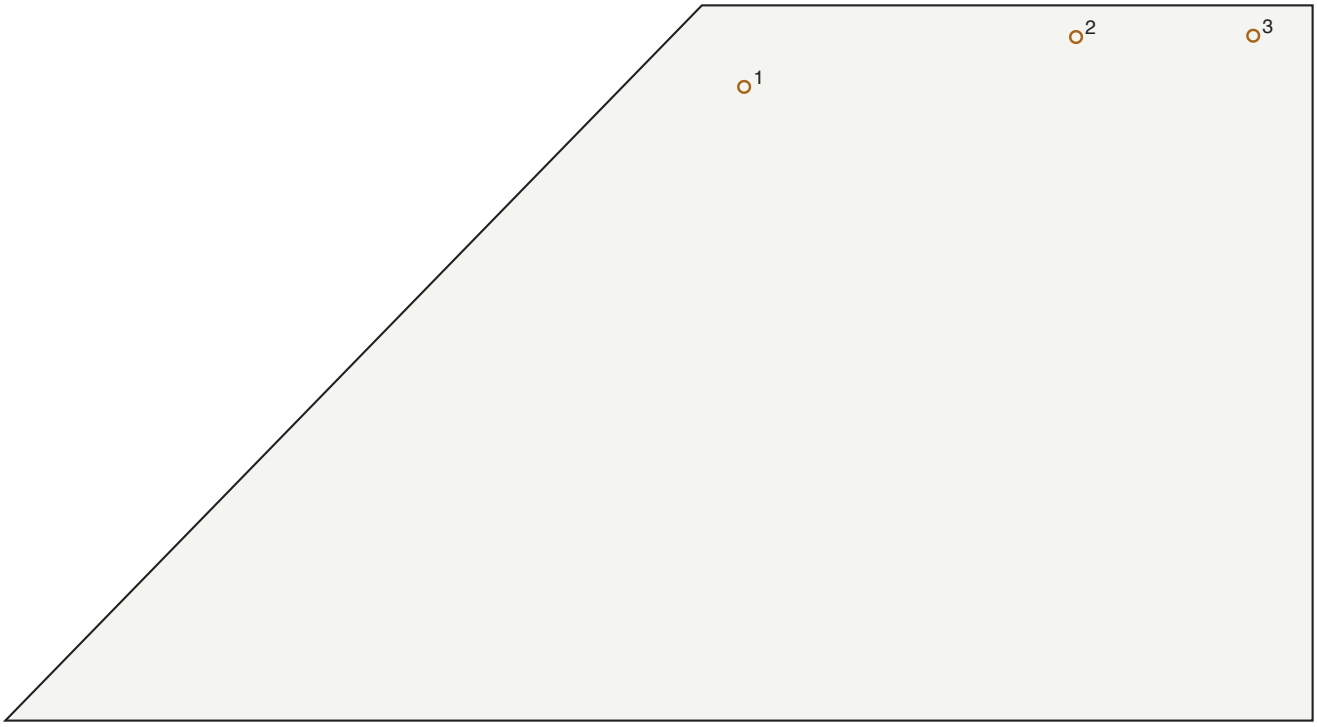
Upright floor (x2)



#	x	y	Hole	For
1	121	56	4.5	toe-filler
2	79	75	4.5	toe-filler
3	131	136	3.2	foot-gasket-groove
4	99	146	16.3	5/8 core rod
5	20	170	4.5	cleat
6	178	170	4.5	cleat
7	67	197	3.2	foot-gasket-groove
8	99	246	16.3	5/8 core rod
9	67	266	3.2	foot-gasket-groove
10	131	266	3.2	foot-gasket-groove
11	20	333	4.5	cleat
12	67	336	3.2	foot-gasket-groove
13	131	336	3.2	foot-gasket-groove
14	178	333	4.5	cleat
15	99	371	63.5	hole saw
16	131	414	3.2	foot-gasket-groove
17	79	426	16.3	5/8 core rod
18	99	426	13.1	1/2-13 bolt, base dowel hole
19	119	426	16.3	5/8 core rod

#	x	y	Hole	For
20	67	461	3.2	foot-gasket-groove
21	99	481	63.5	hole saw
22	20	495	4.5	cleat
23	178	495	4.5	cleat
24	99	555	4.5	cleat

Upright side plate, outer (x2)

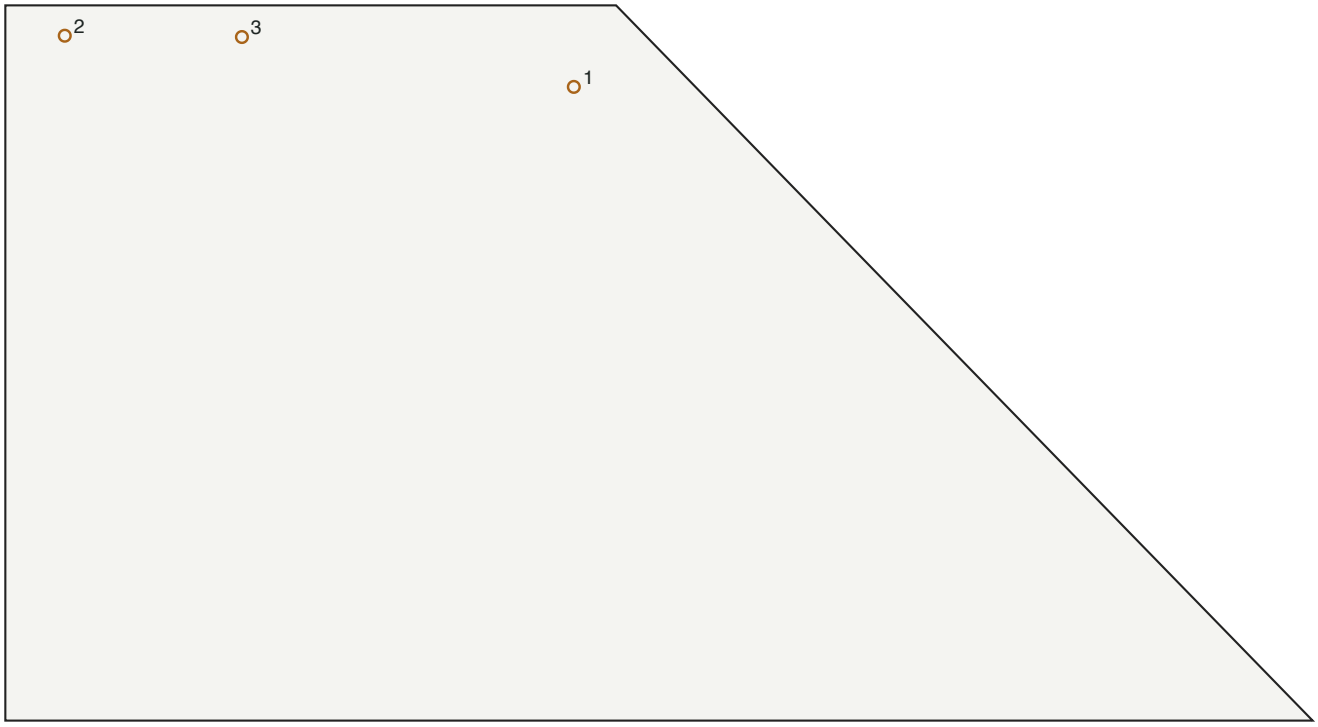


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	280	240	4.5	corner bracket
2	406	259	4.5	saddle
3	473	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (495, 0), (495, 271), (264, 271)

Upright side plate, inner (x2)

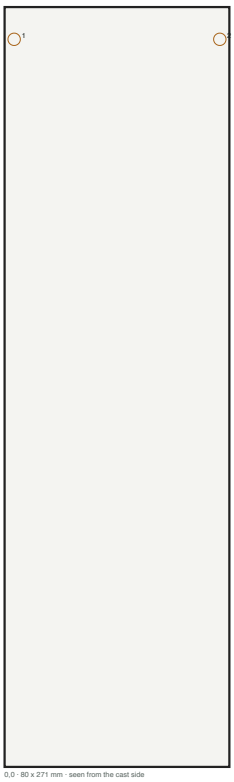


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	215	240	4.5	corner bracket
2	22	260	4.5	corner bracket
3	90	259	4.5	saddle

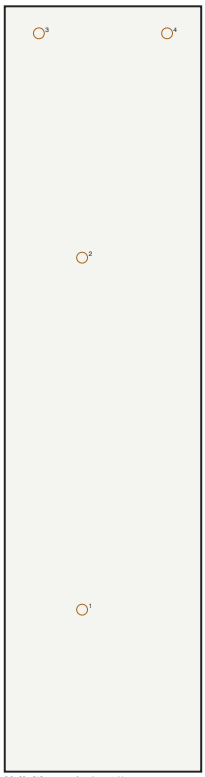
Outline corners (x, y): (495, 0), (0, 0), (0, 271), (231, 271)

Upright back board (x2)



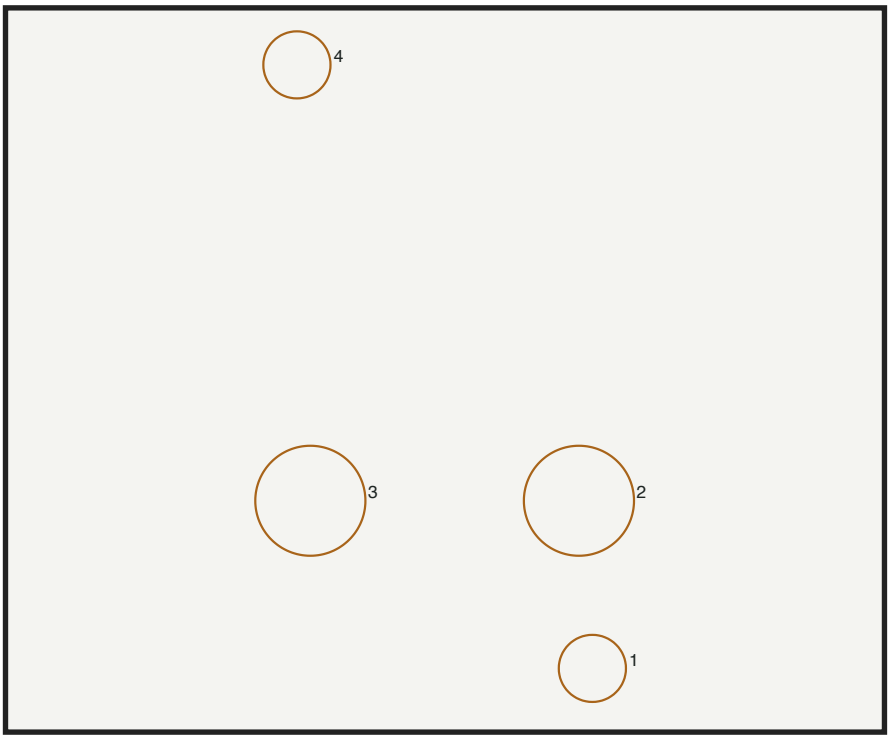
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board (x2)



#	x	y	Hole	For
1	32	66	4.5	slope-plug-lower
2	32	209	4.5	slope-plug-upper
3	14	301	4.5	corner bracket
4	66	301	4.5	corner bracket

Upright bridge (x2)



0,0 · 131 x 108 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	10	10	3/8-16 bolt into the top plate
2	85	34	16.3	5/8 core rod
3	45	34	16.3	5/8 core rod
4	43	100	10	3/8-16 bolt into the top plate

Upright cleat, outer (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

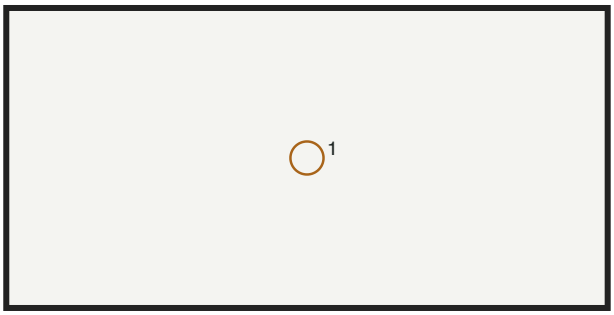
Upright cleat, inner (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

Upright cleat, back (x2)

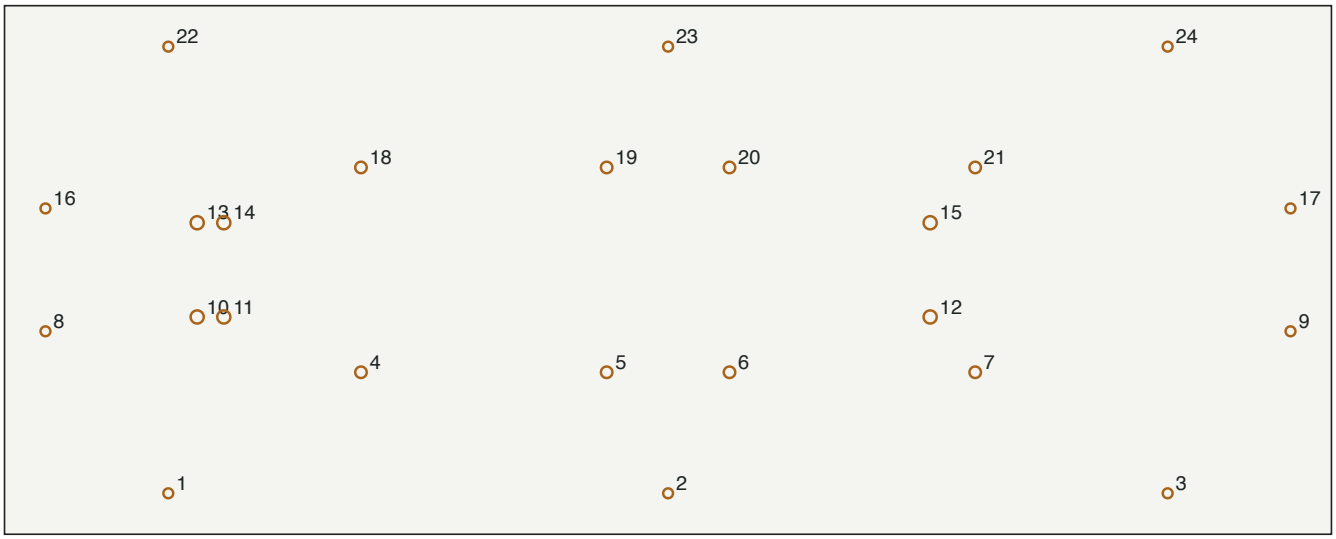


0,0 · 80 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	20	4.5	cleat

Beam

Beam floor



0,0 · 648 x 258 mm · seen from the cast side

#	x	y	Hole	For
1	80	20	4.5	cleat
2	324	20	4.5	cleat
3	568	20	4.5	cleat
4	174	79	5.5	M5 insert screw
5	294	79	5.5	M5 insert screw
6	354	79	5.5	M5 insert screw
7	474	79	5.5	M5 insert screw
8	20	99	4.5	cleat
9	628	99	4.5	cleat
10	94	106	6.5	M6 insert screw
11	107	106	6.5	M6 insert screw
12	452	106	6.5	M6 insert screw
13	94	152	6.5	M6 insert screw
14	107	152	6.5	M6 insert screw
15	452	152	6.5	M6 insert screw
16	20	159	4.5	cleat
17	628	159	4.5	cleat
18	174	179	5.5	M5 insert screw
19	294	179	5.5	M5 insert screw
20	354	179	5.5	M5 insert screw
21	474	179	5.5	M5 insert screw
22	80	238	4.5	cleat
23	324	238	4.5	cleat
24	568	238	4.5	cleat

Beam top wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	37	115	13.1	1/2 core rod
6	81	115	13.1	1/2 core rod
7	487	115	13.1	1/2 core rod
8	531	115	13.1	1/2 core rod
9	84	133	4.5	back-edge
10	219	133	4.5	back-edge
11	351	133	4.5	back-edge
12	484	133	4.5	back-edge
13	33	154	4.5	corner bracket
14	535	154	4.5	corner bracket

Beam bottom wall

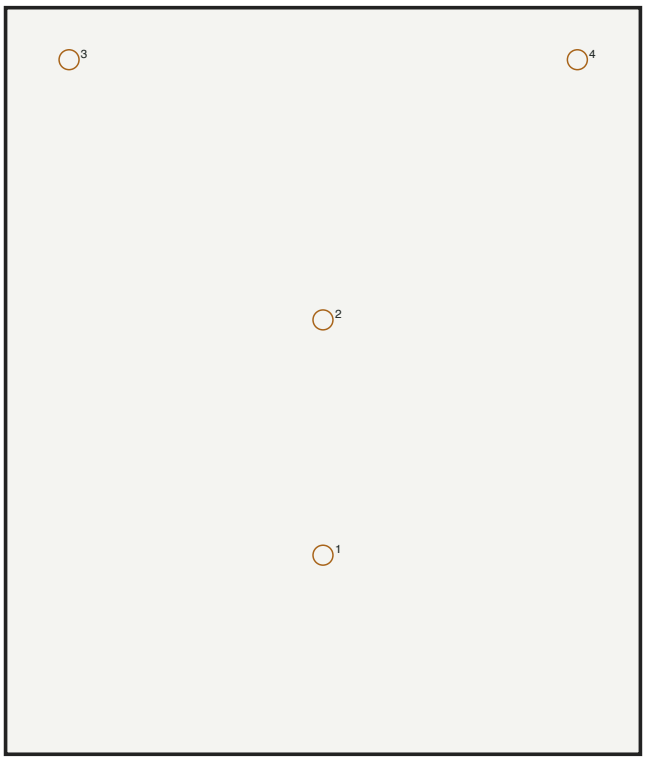


0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	509	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod

#	x	y	Hole	For
9	487	115	13.1	1/2 core rod
10	531	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	535	154	4.5	corner bracket

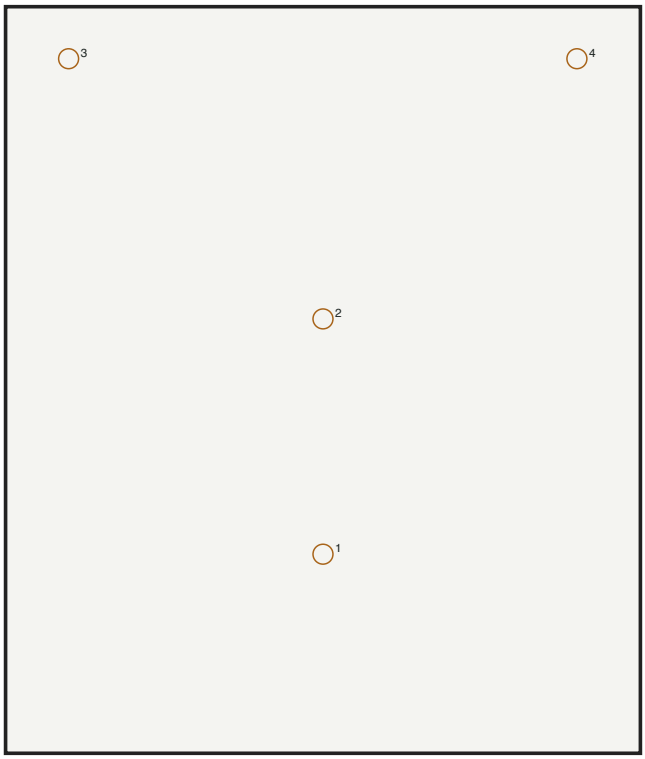
Beam end wall, left



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam end wall, right



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam cleat, top



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, bottom



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, left



0,0 · 140 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat

Beam cleat, right



0,0 · 140 x 40 mm · seen from the outside (top)

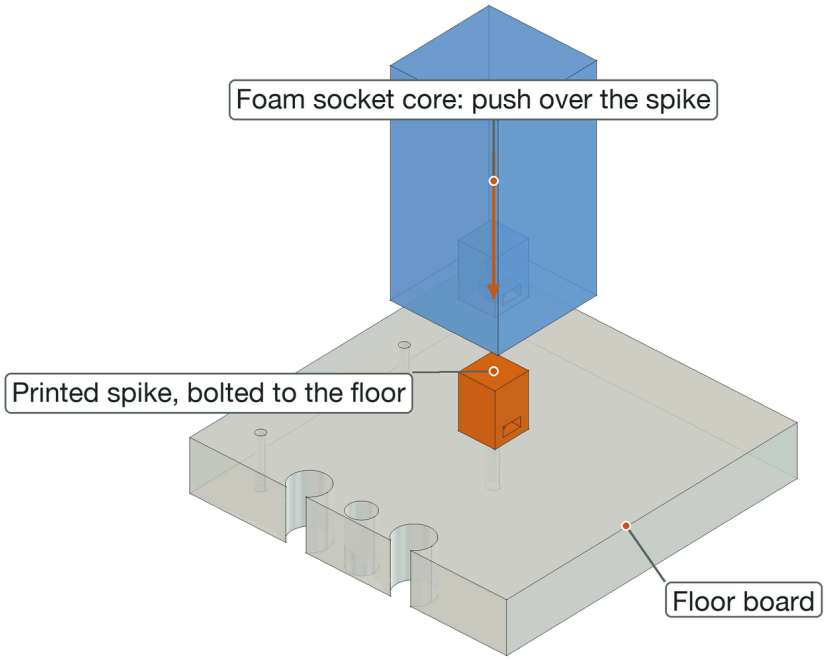
#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat

6. Cut the foam

Cut with a fine saw, a hot wire, or a long sharp knife and a sanding block. Wrap each piece in packing tape (smooth, no wrinkles: the tape is the cast face). For the 90 mm cores glue two layers of 2 in foam with a foam-safe adhesive (not solvent cement) or double-sided tape, then trim. Make each pocket core's post hole a snug 12 mm square, 30 mm deep, and glue the printed post in.

Casting	Piece	Qty	Size and how
base-front	center	1	XPS block 100 x 185 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	center	1	XPS block 100 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-L	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-L	1	XPS block 37 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-L	1	XPS block 68 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 68 x 100 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-c-L	1	XPS block 80 x 63 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 50 x 183 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-L	1	XPS block 35 x 43 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-L	1	XPS block 35 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-L	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-R	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-R	1	XPS block 37 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-R	1	XPS block 68 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-R	1	XPS block 68 x 100 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-c-R	1	XPS block 80 x 63 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.

Casting	Piece	Qty	Size and how
base-rear	knee-R	1	XPS block 50 x 183 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-R	1	XPS block 35 x 43 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-R	1	XPS block 35 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-R	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	socket	4	XPS block 41 x 41 x 80 mm, taped; dig it out after curing (a little acetone dissolves the rest).



7. Prepare the surfaces

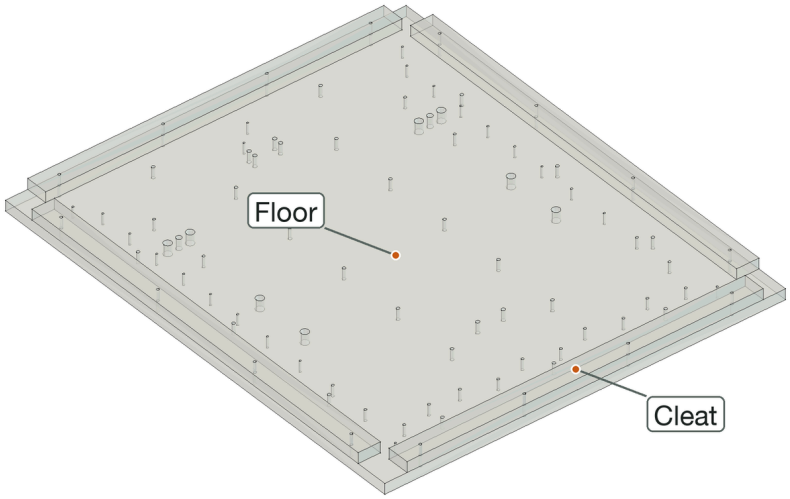
- Wax every face the mix touches, boards and removable printed pieces alike: 3-5 thin coats of paste wax, each buffed, then a coat of PVA. Don't wax the pieces that stay in (core posts, cast-in parts): scuff them and wipe them with acetone so the epoxy grips.
- The castings' outside edges are rounded 10 mm: glue 3/8 in quarter-round molding into the inside corners that form outside edges (floor to wall), and run a small silicone fillet into the others.
- Grease every screw, bolt and rod that will come out of the casting.

8. Put the molds together

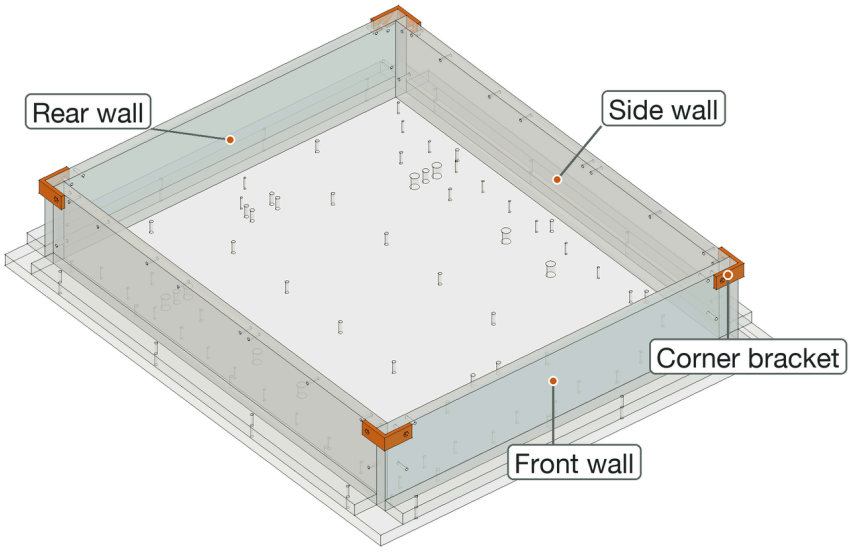
Base, rear half (cast first)

Both base halves are cast on the same floor board. The rear half goes first, against a partition at the split; then the partition comes out and the front half is cast against the rear half's waxed split face. The two top faces come out flush, the split faces match, and the front tubes slide straight onto the rear half's spigots. The base is cast upside down (its top face on the floor); these pictures show the mold the right way up.

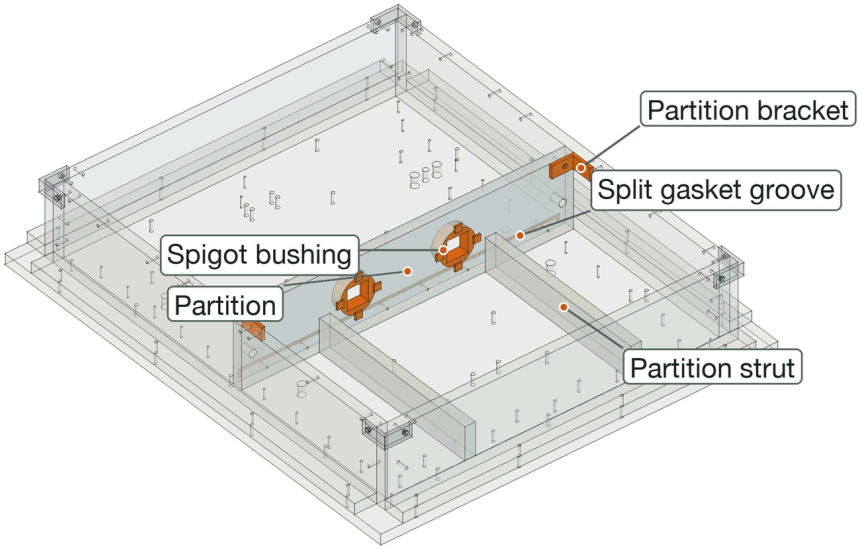
1. Stand the floor on three battens (the insert screws and floor bolts come up from underneath) and bolt the cleats down along its edges.



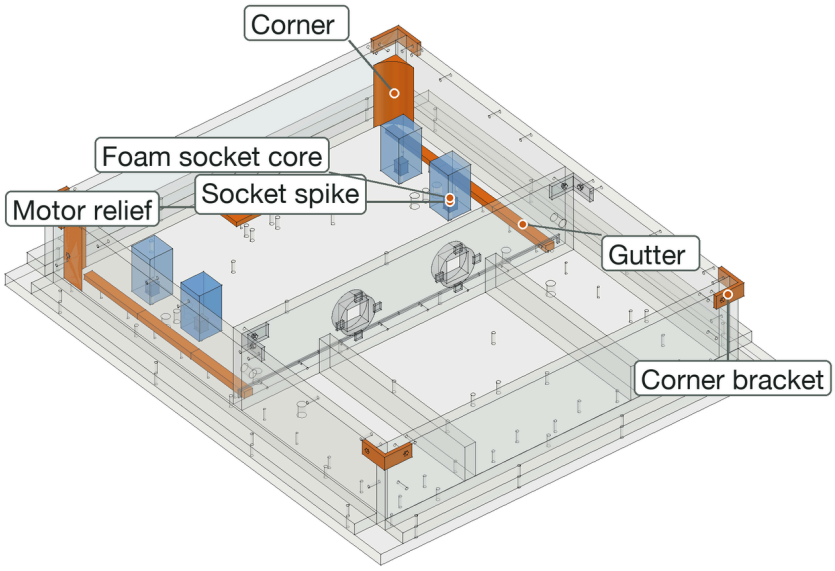
2. Stand the walls on the floor against the cleats and fit the four corner brackets at the top corners. Measure the diagonals before you tighten: they should match within 2 mm.



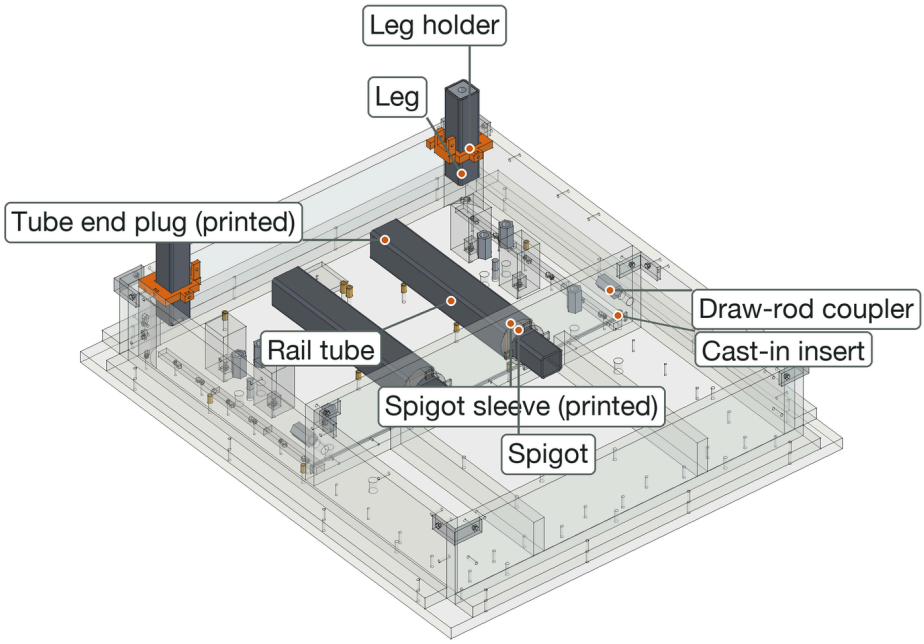
3. Fit the partition between the side walls with its two brackets on the front side, and the two struts from the front wall to its foot. Push the gasket-groove beads onto its pins and the spigot bushings into its holes from the front.



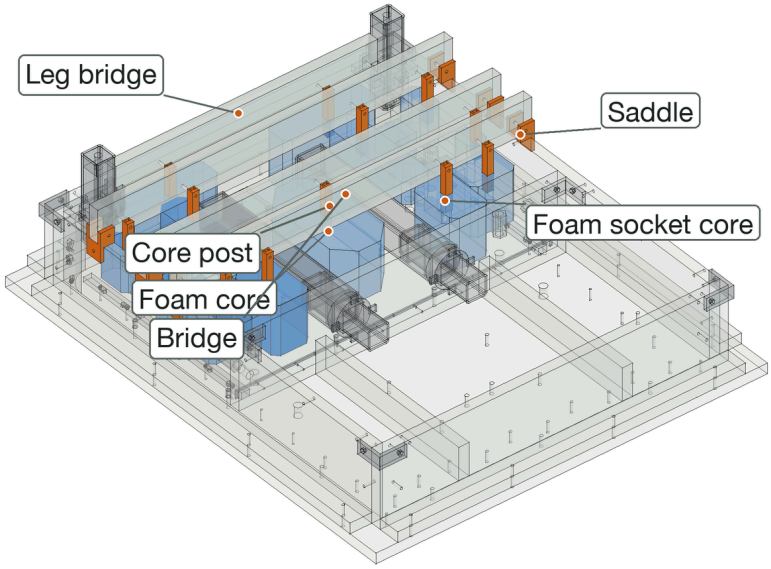
4. Bolt on the gutter formers, the motor relief (pins) and the rear corner fillers; bolt the socket spikes to the floor and push a taped foam block over each.



5. Screw the inserts to the floor and the tie-rod couplers onto their greased bolts; hold the draw-rod couplers on bolts through the partition. Slide the spigots through the bushings and put the rear tubes on them, with their sleeves and end plugs. Bolt the leg holders into the corners, drop the legs in, set each at its height and tighten its pinch screw.

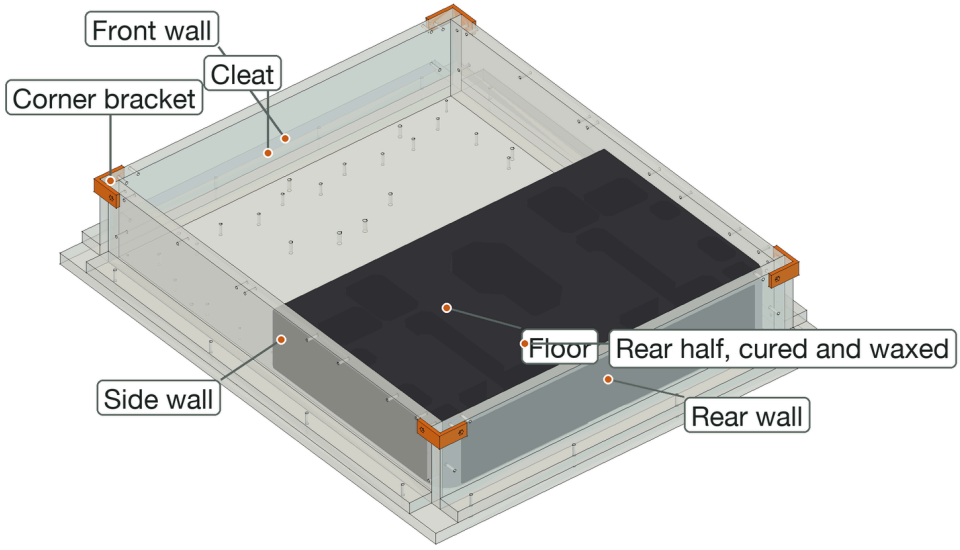


6. Cast the 20 mm skin first. Then set the foam cores on it, drop the bridges into the saddles and bolt each core's post to its bridge. Fill to the pour line.

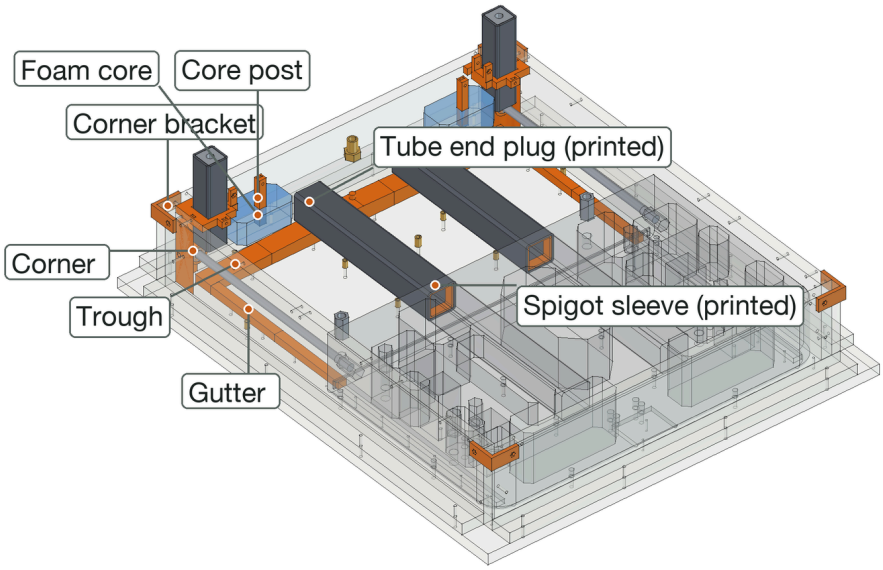


Base, front half (against the rear half)

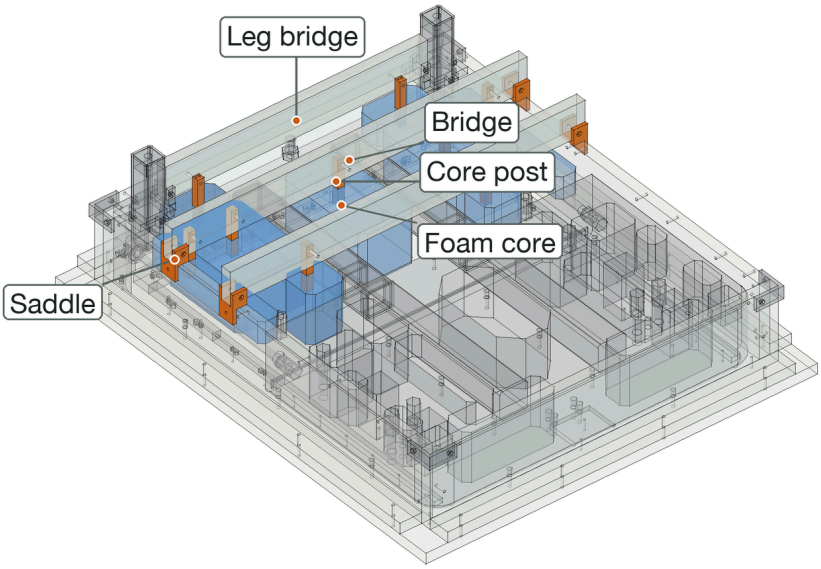
1. After the cure: unbolt the bridges, the partition and its struts, and leave the rest. Dig the foam out of the four sockets, pull the gasket-groove beads, tape over the groove, and wax and PVA the rear half's split face and spigots.



2. Bolt on the trough and gutter formers (the drain liner on the trough's spigot) and the front corner fillers; put the front tubes, sleeves and plugs onto the spigots; screw the draw rods into the rear half's couplers with their ends in the corner fillers; inserts, couplers and legs as before.



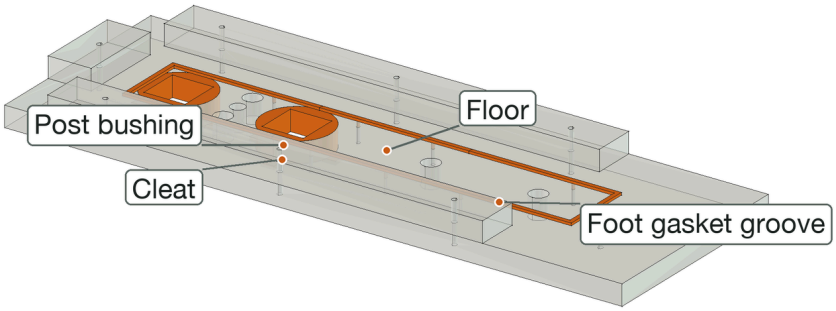
3. Skin, foam cores, bridges, fill. After the cure unbolt everything and split the halves with a few wedges at the split, gently.



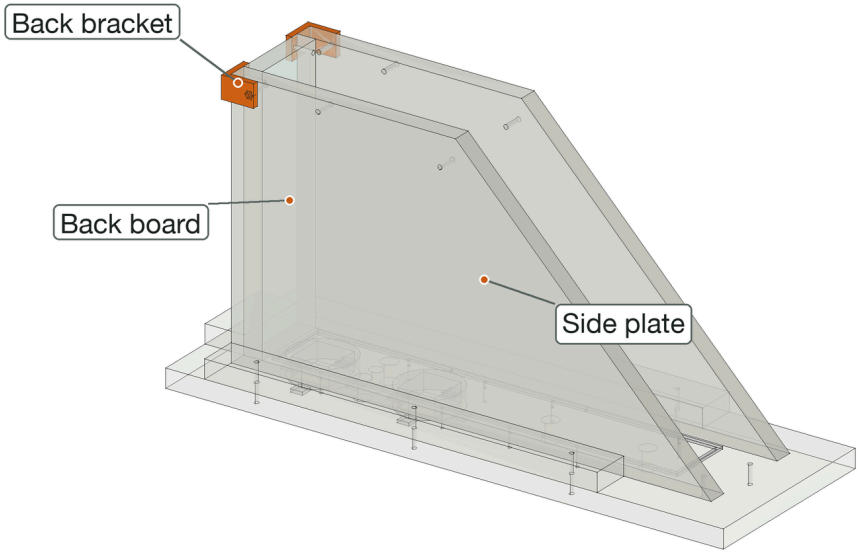
Uprights (two molds, cast together)

The uprights are cast foot down. Both are the same, so build two of everything.

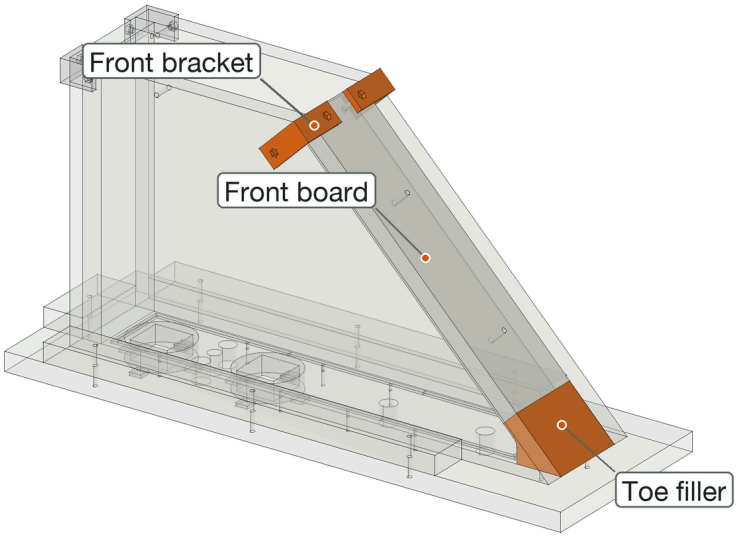
1. Push the post bushings into the floor from below and pin the gasket-groove bead to it. Bolt the cleats down.



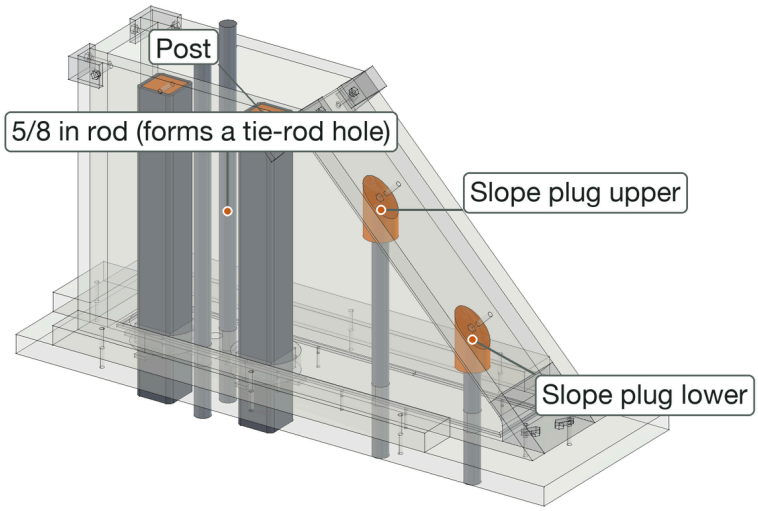
2. Stand the side plates against their cleats and the back board between them; fit the two back brackets.



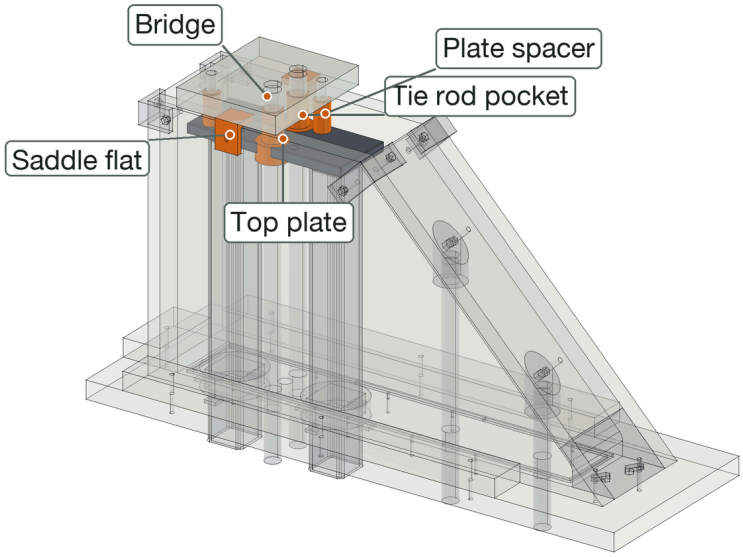
3. Bolt the toe filler to the floor between the side plates, set the front board's foot in its lip and fit the two front brackets at the top.

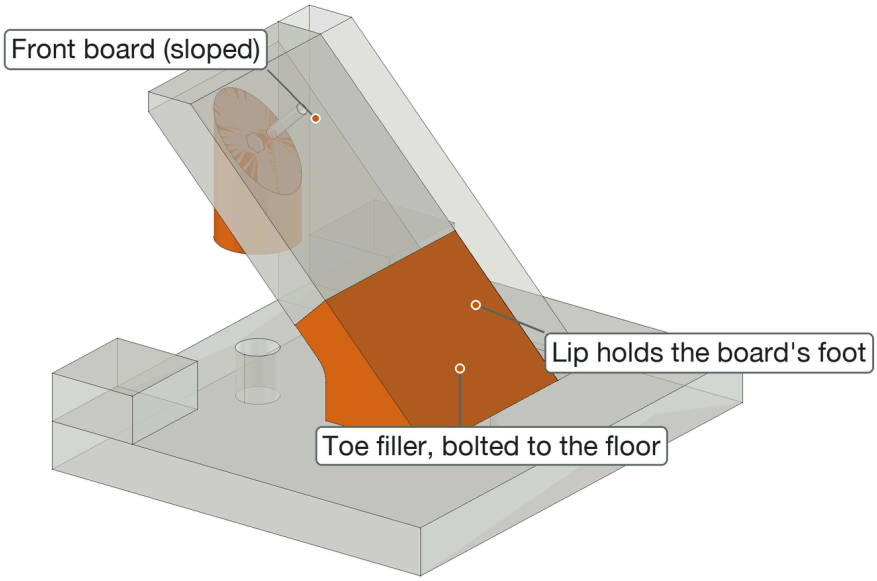


4. Stand the mold on its four 2x4 blocks. Push the posts up through the bushings onto the bench, and the greased 5/8 in rods up through the floor: the two short ones into the slope plugs (bolted to the front board), the two long ones on through the top. The dowel-hole bolt and nut go in from underneath.



5. Fill to the line and vibrate. Then slide the top pocket plugs down the long rods, hang the top plate from the bridge on its greased 3/8 bolts and spacers, and drop the bridge onto its saddles.

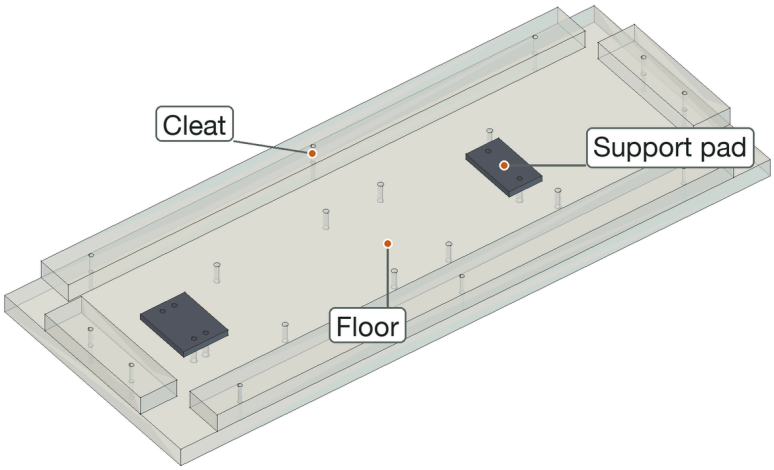




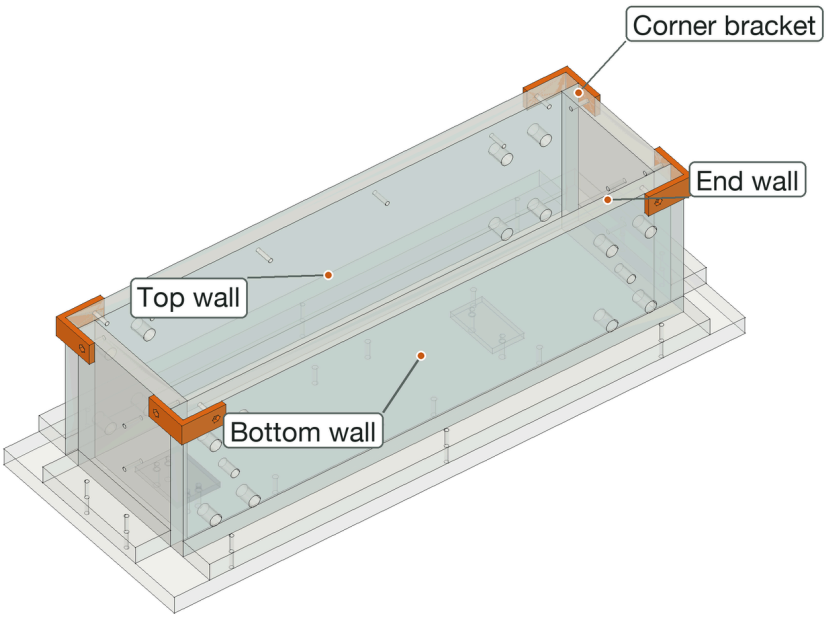
Beam

The beam is cast with its front (rail) face down on the floor and its back open.

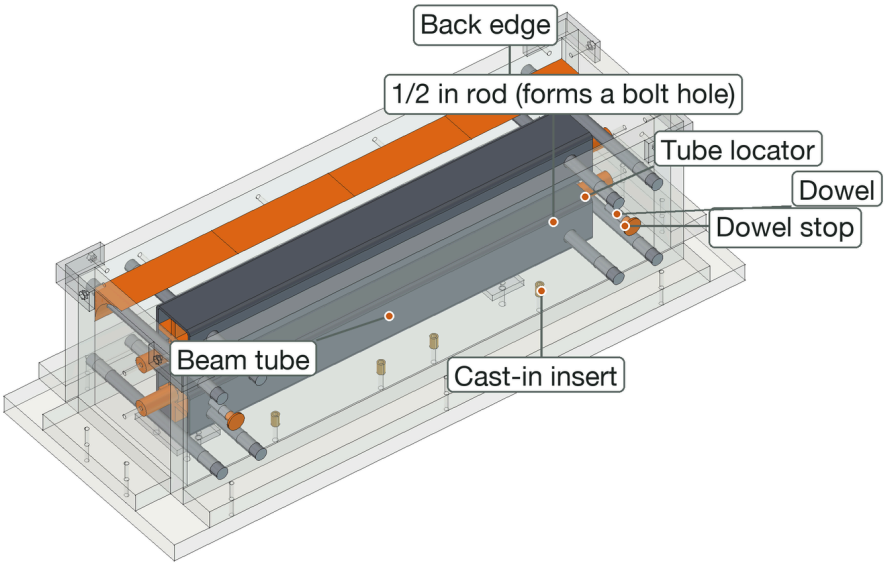
1. Bolt the cleats down; screw the support pads and riser inserts to the floor on greased screws.

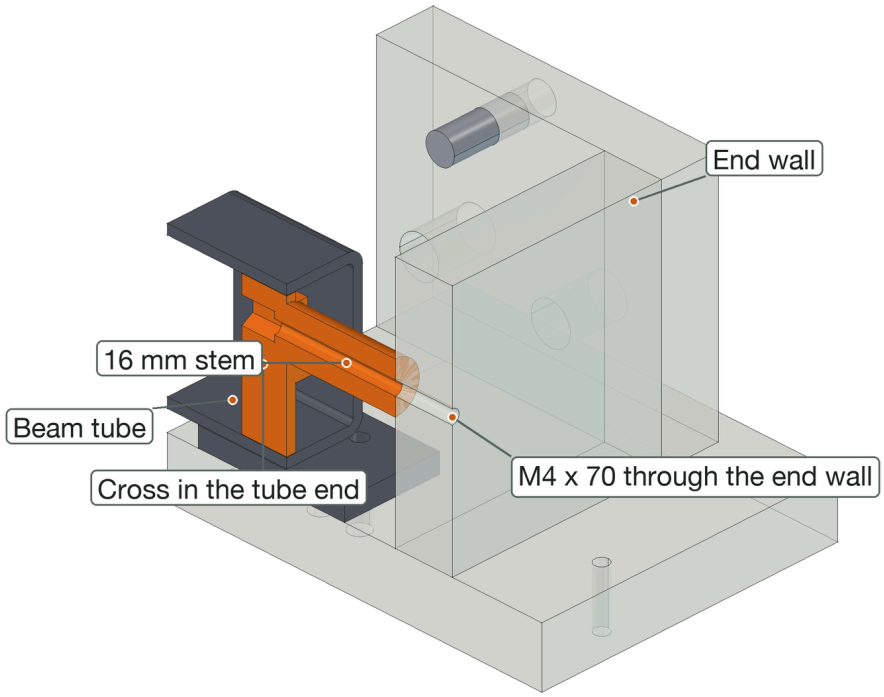


2. Stand the walls against the cleats and fit the four corner brackets.



3. Bolt the tube locators through the end walls (M4 x 70 into the nut in each cross) and hang the tubes between them. Bolt the back-edge fillers to the top wall, push the dowels into the bottom wall onto their printed stops, and slide the greased 1/2 in rods through both walls for the eight bolt holes. Fill to the line.





9. Pouring, and the pour line

- The pour line is 25 mm below the wall tops. Mark it inside each wall with a pencil line, and screed the top with a straight board that has two 25 mm feet riding on the wall tops.
- Vibrate each layer until it stops bubbling; the brackets and cleats hold the walls, so you can vibrate the floor board hard. Check the corner brackets and saddle bolts are still tight halfway through.
- Keep the mix off the bridges and saddles; scrape it off before it cures.

10. Taking it apart, and reuse

1. Leave it the full cure time (build guide). Unbolt the bridges and saddles, then the corner brackets, then the cleats; pull the walls straight off (a plastic wedge in the corner helps).
2. Unscrew every insert screw, coupler bolt and rod from outside, then lift the casting off the floor (or the floor off the casting).
3. Unbolt and pull the formers and fillers (the bolts come out from the board side first). A wooden dowel and a light tap frees a stuck one. Dig out the socket foam; the pocket cores can stay.
4. Scrape the boards, re-wax them, and store the kit pieces with their nuts in a box per mold. The same kit does the next casting; only the formers and pieces that stay in are per casting.

11. If something doesn't fit

Problem	Fix
A nut won't slide under the hole	Clean the pocket with a 7 mm nut on a bolt; a warm nut (hair dryer) pushes in

Problem	Fix
A bracket or saddle is loose on the board	Your board is thinner than set: shim with tape, or set `mold_board` and reprint
A pin won't go into its hole	Run a 1/8 in bit through the hole again; trim the pin tip with a knife
A foam core moves while pouring	Check its post bolt is tight and the post is glued; add carpet tape under it
Mix leaks at a corner	A silicone fillet inside the corner before waxing; check the diagonals and brackets
A former is stuck after curing	Bolts out first; then a dowel and a light tap from its thick end